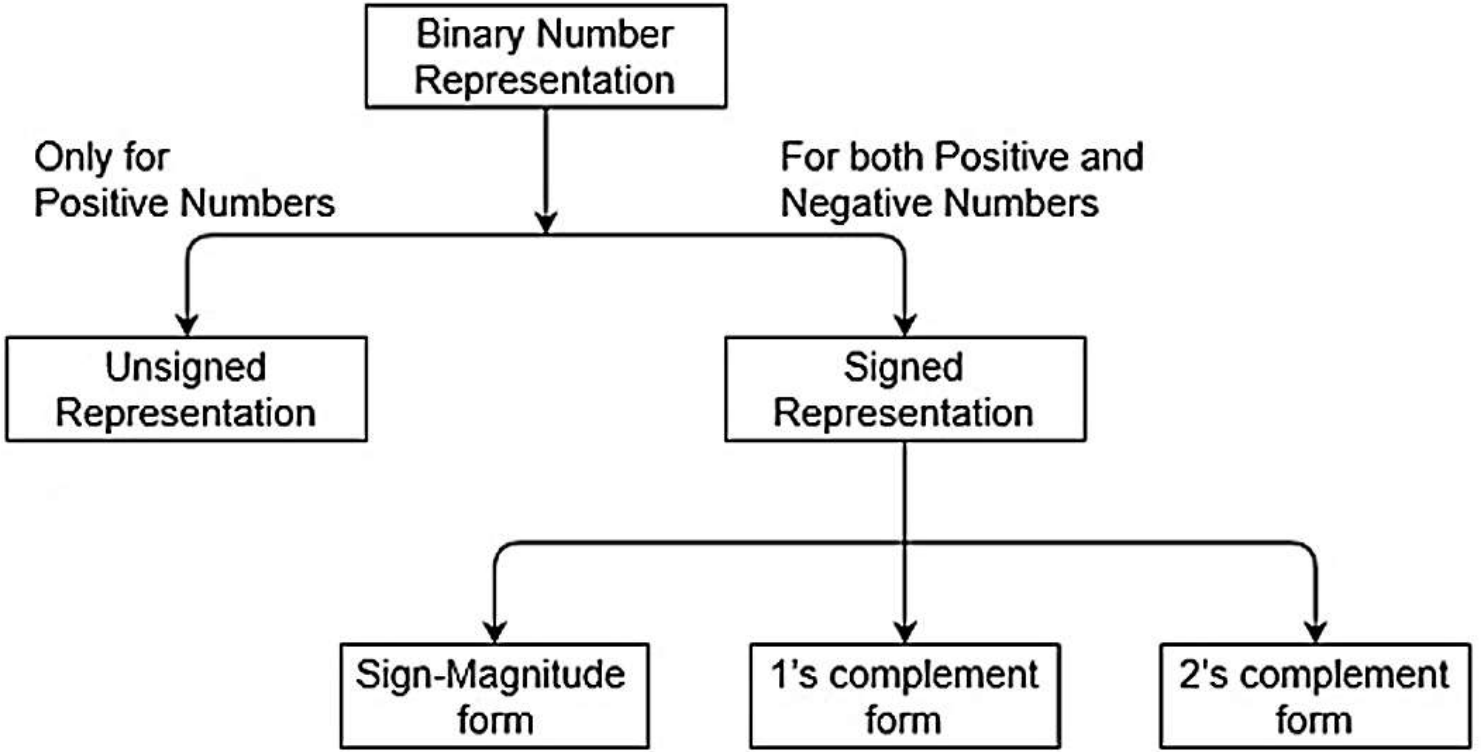
Data Representation & Storing

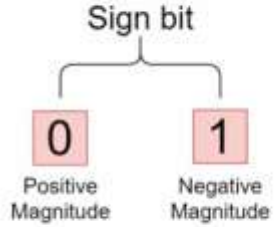

Binary Number Representation
Only for Positive Numbers
For both Positive and Negative Numbers
Unsigned Representation
Signed Representation
Sign-Magnitude form
1's complement form
2's complement form

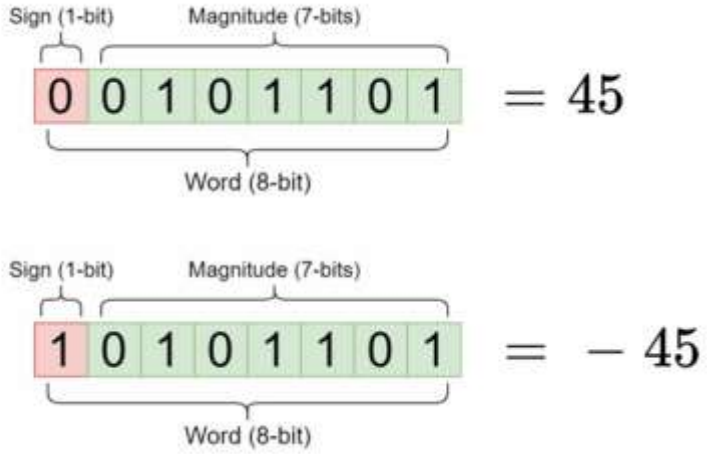
1. Sign-and-Magnitude

- The easiest representation
- The leftmost bit represents the sign of the number.

  0 if positive & 1 if negative

- Try to solve : Convert in (8-bit):
  - $(11111100)_2$
  - $(01110100)_2$
  - $(-55)_{10}$
  - $(60)_{10}$





Example

1. $(+7)_{10} = (00000111)_2$

   $(-7)_{10} = (10000011)_2$

2. $(-10)_{10} = (10001010)_2$

   $(+10)_{10} = (00001010)_2$

2. One's Complement

- Positive numbers are the same as in sign-and magnitude

Example: $(+5)_{10} = (00000101)_2$ (8 bit)

- For negative numbers, their representations are obtained by changing bit $0 \rightarrow 1$ & $1 \rightarrow 0$
- Example:

$(+5)_{10} = (0\ 0\ 0\ 0\ 0\ 1\ 0\ 1)_2$

$(-5)_{10} = (1\ 1\ 1\ 1\ 1\ 0\ 1\ 0)_2$ (8 bit)

- Try to solve: Convert in (8-bit):
  - $(-40)_{10}$
  - $(-112)_{10}$
  - $(120)_{10}$

Example

Get the representation of one's complement (6-bit) for the following numbers:

1. $(+7)_{10}$

   Solution:

   $(+7)_{10} = (0\,0\,0\,1\,1\,1)_2$

2. $(-10)_{10}$

   Solution:

   $(+10)_{10} = (0\,0\,1\,0\,1\,0)_2$

   So,

   $(-10)_{10} = (1\,1\,0\,1\,0\,1)_2$

3. Two's Complement

- Representation of its negative number is obtained by adding 1 to the one's complement of the number.

Example: Convert (–5) into two's complement representation in 8 bits.

Solution:

- First, obtain (+5) representation in 8 bits → $(00000101)_2$
- Obtain one's complement for (–5) → $(11111010)_2$

Add 1 to the one's complement number:

$$\begin{array}{r} (11111010)_2 \\ 1+ \\ \hline (11111011)_2 \end{array}$$

–5 in two's complement is $(11111011)_2$

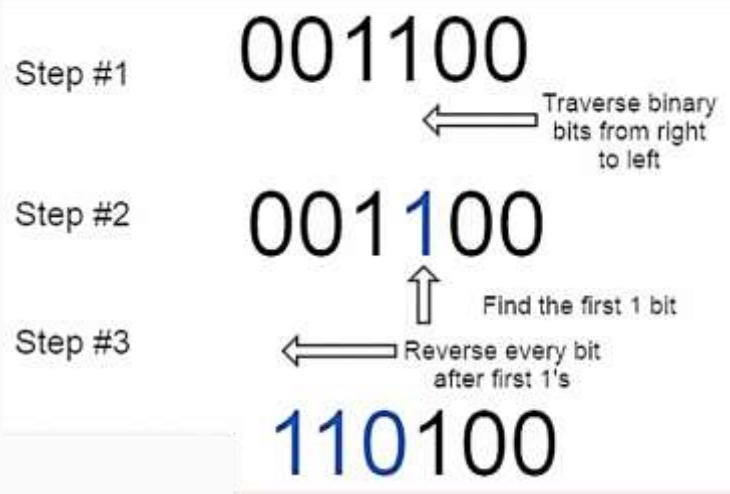
Two’s Complement (Shortcut)

- Another way to get the two’s complement is the copy and complement method.
- In this method you copy numbers from the right till the first one and then you reverse (complement the rest)

Example:

- $(+12)_{10}$ is 0 0 1 1 0 0
- $(-12)_{10}$ in two’s complement: 1 1 0 1 0 0

- Try to solve :
  - $(-10)_{10}$
  - $(88)_{10}$
  - $(-110)_{10}$



Thanks!